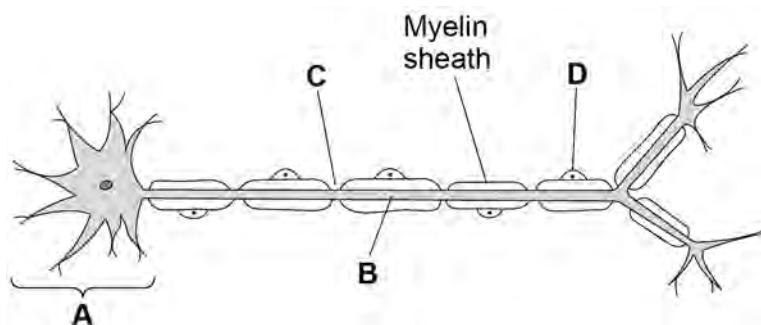


0 6 . 1

Figure 2 shows a drawing of a myelinated motor neurone.

Figure 2



Name the parts labelled **A** to **D**.

[2 marks]

A _____

B _____

C _____

D _____

0 6 . 2

Osmotic demyelination syndrome (ODS) results in destruction of the myelin sheath of neurones in the brain.

ODS can **increase** reflex response times.

Explain why.

[3 marks]

0	6	.	3
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The parasympathetic nervous system (PNS) uses acetylcholine as its neurotransmitter. Atropine is an anticholinergic drug that attaches to postsynaptic receptors.

Atropine can be used to increase the heart rate in patients with bradycardia (a slower than normal heart rate).

Use the information provided to explain how atropine causes an increase in the heart rate.

[3 marks]

Question 6 continues on the next page

0 6 . 4

One study has suggested that the atropine dosage for severely obese male patients should be 0.01 mg kg^{-1} of lean body weight (LBW).

LBW is determined by using this equation.

$$\text{LBW} = \frac{9.27 \times 10^3 \times \text{TBW}}{(6.68 \times 10^3) + (216 \times \text{BMI})}$$

where TBW = total body weight

BMI = body mass index

Calculate the suggested dosage of atropine, in mg, for a severely obese male patient whose:

- TBW = 115.2 kg
- BMI = 41

Give your answer to 3 significant figures.

Show your working.

[2 marks]

Answer _____ mg

1

Turn over for the next question

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ANSWER IN THE SPACES PROVIDED**

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